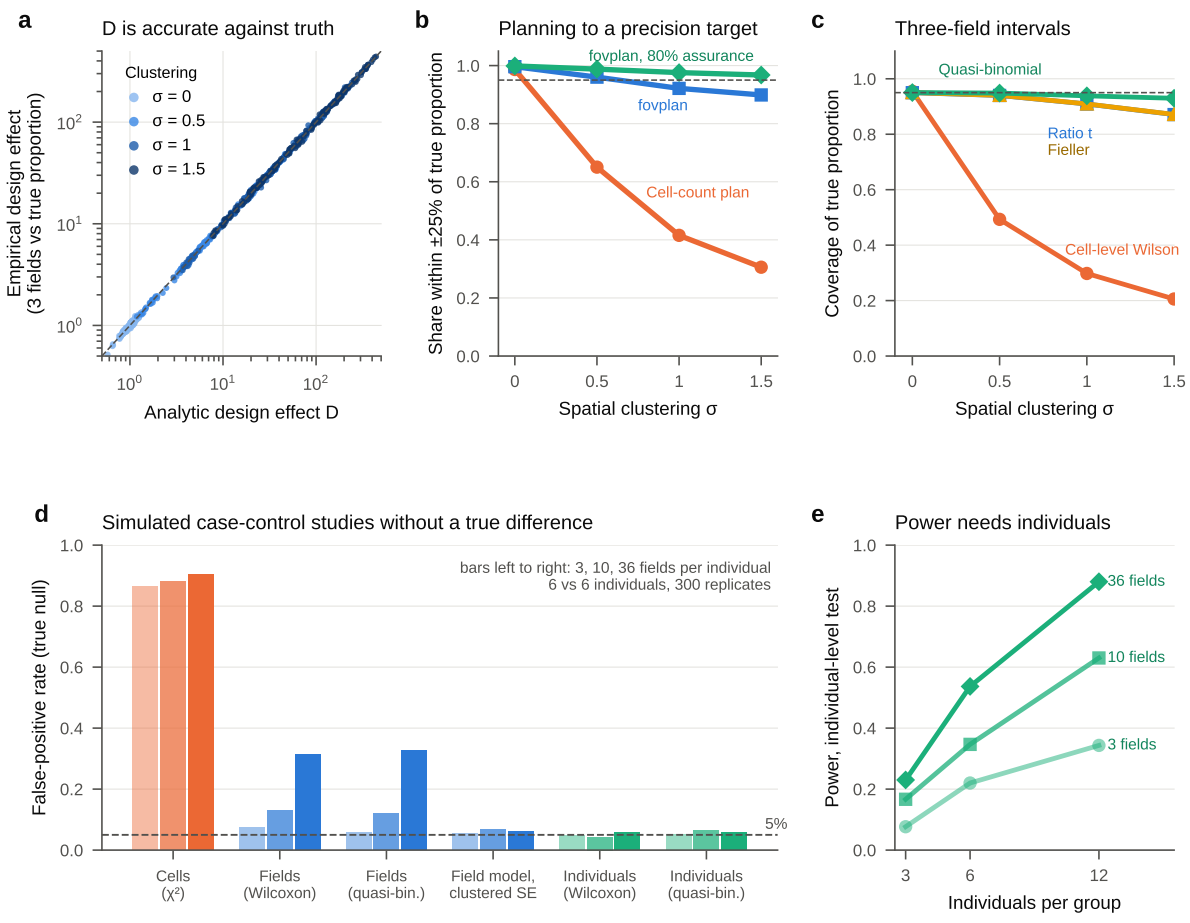
a
D is accurate against truth
Clustering
σ = 0
σ = 0.5
σ = 1
σ = 1.5
Empirical design effect
(3 fields vs true proportion)
Analytic design effect D
b
Planning to a precision target
fovplan, 80% assurance
fovplan
Cell-count plan
Share within ±25% of true proportion
Spatial clustering σ
c
Three-field intervals
Quasi-binomial
Ratio t
Fieller
Cell-level Wilson
Coverage of true proportion
Spatial clustering σ
d
Simulated case-control studies without a true difference
bars left to right: 3, 10, 36 fields per individual
6 vs 6 individuals, 300 replicates
5%
False-positive rate (true null)
Cells (χ²)
Fields (Wilcoxon)
Fields (quasi-bin.)
Field model, clustered SE
Individuals (Wilcoxon)
Individuals (quasi-bin.)
e
Power needs individuals
36 fields
10 fields
3 fields
Power, individual-level test
Individuals per group